

L12 : SVM (contd.)

- Agenda:
 - Soft margin SVM
 - Kernel methods

Recall:

For linearly separable training data, max-margin classifier is:

$$\begin{aligned} \min_{w, b} \quad & \frac{1}{2} \|w\|^2 \\ \text{s.t.} \quad & y_i (w^T x_i + b) \geq 1 \quad \forall i \end{aligned}$$

For non-separable data, we relax this as follows:

$$\begin{aligned} \min_{w, b, \theta} \quad & \frac{1}{2} \|w\|^2 + C \sum_i^{\frac{n}{d}} \theta_i^d \\ \text{s.t.} \quad & y_i (w^T x_i + b) \geq 1 - \theta_i \quad \forall i \\ & \theta_i \geq 0 \end{aligned}$$

Here, $d \geq 1$ (retaining convexity of objective), $C > 0$.

$\theta_i > 0 \Rightarrow$ data point i is an outlier.

- $\theta \in (0, 1) \Rightarrow$ point is still on the "correct" side of the separating hyperplane
- $\theta > 1 \Rightarrow$ point is misclassified by the model

Assume $p = 1$ for simplicity.

Exercise: Write KKT conditions for this formulation.
formulate the dual.

Dual:

$$\max_{\alpha \in \mathbb{R}^n} \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j \langle x_i, x_j \rangle$$

$$\text{s.t.} \quad \sum \alpha_i y_i = 0 \\ 0 \leq \alpha_i \leq C \quad \forall i$$

(assuming $p=1$)

★ Solving the dual only involves training inputs via inner products $\langle x_i, x_j \rangle$.

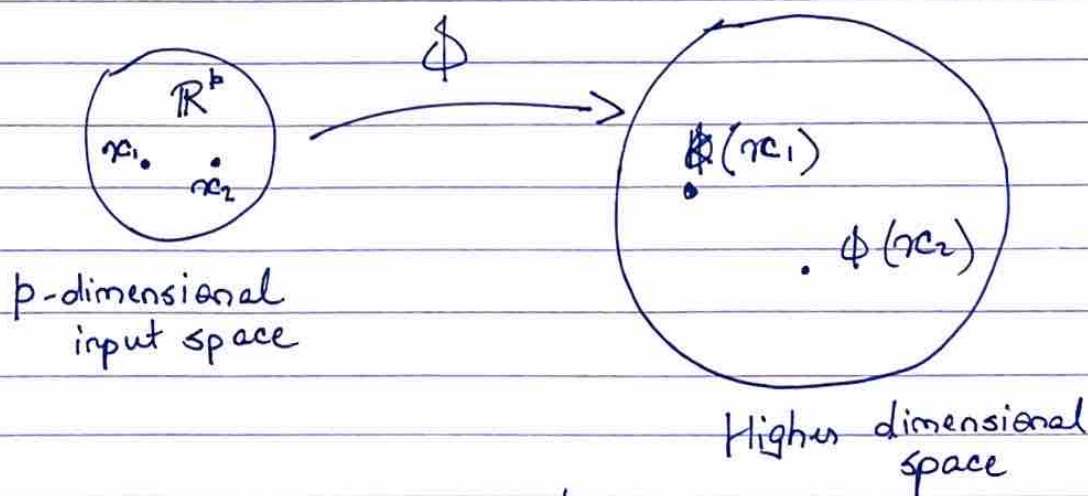
★ Influence on a test input x only requires evaluation of $\langle x_i, x \rangle$ for support vectors x_i .

↓
these inputs for which $\alpha_i^* > 0$.

These properties are crucial when we "kernelize" SVMs.

Kernel methods

- Widely used in ML
- Can be used to extend algorithms like SVM & regression into non-linear methods!
Eg: Kernel SVM enables a non-linear separation boundary.
- Key idea: Kernel functions implicitly define an inner product in a higher dimensional space.



$$K(x_1, x_2) = \langle \phi(x_1), \phi(x_2) \rangle.$$

- Training data may not be linearly separable in "original" input space, but may be linearly separable in the higher dimensional space post transformation by ϕ .

↳ ^{SVM} Kernel ~~methods~~ allows us to find a separating hyperplane in this higher dimensional space! (non linear separation boundary in original space).

Formally, a kernel function is a mapping

$$K: \mathbb{R}^p \times \mathbb{R}^p \rightarrow \mathbb{R}.$$

Def: A kernel is said to be positive definite symmetric (PDS) if, for any $x_1, \dots, x_m \in \mathbb{R}^p$

$$K = [K(x_i, x_j)]_{1 \leq i, j \leq m} \text{ is positive semidefinite}$$

Fact: For any PDS kernel^K, there exists $\phi: \mathbb{R}^p \rightarrow H$
s.t. ↓
higher dim.
Hilbert space

$$K(x_1, x_2) = \langle \phi(x_1), \phi(x_2) \rangle_H.$$

Moral: PDS kernels are "good"!

Eg: Polynomial kernels: For $c > 0$, a polynomial kernel of degree $d \in \mathbb{N}$ is defined as:

$$K(x, x') = (\langle x, x' \rangle + c)^d$$

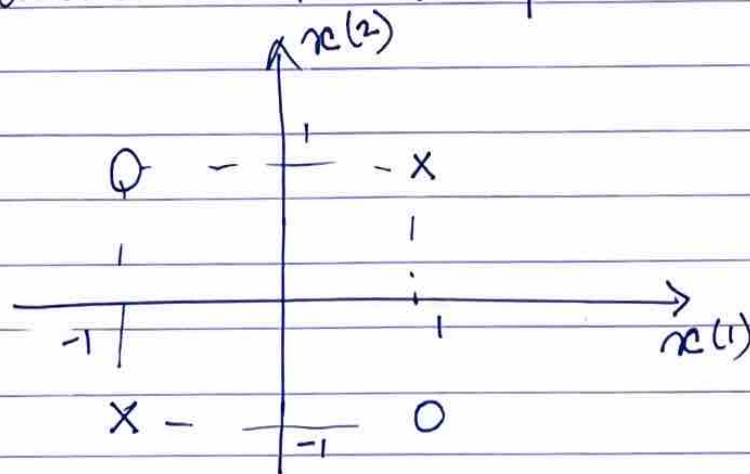
Consider the special case $p = d = 2$.

$$\begin{aligned} K(x, x') &= (x_1 x'_1 + x_2 x'_2 + c)^2 \\ &= x_1^2 x'^2_1 + x_2^2 x'^2_2 + c^2 + 2 x_1 x_2 x'_1 x'_2 \\ &\quad + 2c x_1 x'_1 + 2c x_2 x'_2 \end{aligned}$$

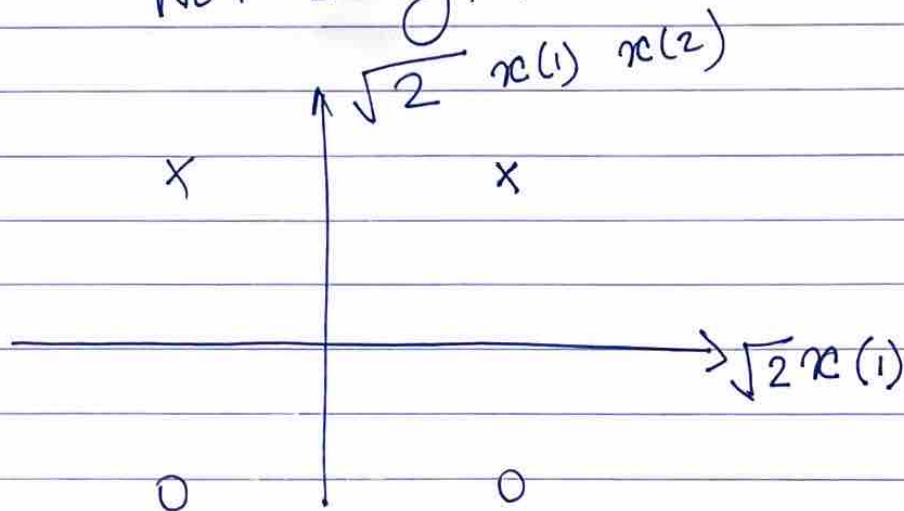
$$= \underbrace{\begin{pmatrix} x_1^2 \\ x_2^2 \\ \sqrt{2} x_1 x_2 \\ \sqrt{2c} x_1 \\ \sqrt{2c} x_2 \end{pmatrix}}_{\phi(x)} \cdot \underbrace{\begin{pmatrix} x_1'^2 \\ x_2'^2 \\ \sqrt{2} x_1' x_2' \\ \sqrt{2c} x_1' \\ \sqrt{2c} x_2' \end{pmatrix}}_{\phi(x')}$$

Moral: 2-Inputs mapped to 6-D space!

Illustration: The XOR problem.



NOT Linearly separable



Separable!

Eg The Gaussian kernel is defined as:

$$K(x, x') = \exp\left(-\frac{\|x - x'\|^2}{2\sigma^2}\right)$$

↳ Defines an inner product in an infinite dimensional space.

Eg Laplacian kernel

$$K(x, x') = \exp(-\lambda \|x - x'\|)$$

Kernel SVM

The dual formulation of SVM is amenable to "Kernelization".

$$\max_{\alpha \in \mathbb{R}^n} \quad \sum_1^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j \underbrace{y_i y_j}_{\text{replaces } \langle x_i, x_j \rangle} K(x_i, x_j)$$

$$\text{s.t.} \quad 0 \leq \alpha_i \leq C \quad \forall i \\ \sum \alpha_i y_i = 0$$

↳ finds a soft margin separating hyperplane in the higher dimensional space with inputs $\{\phi(x_i)\}$.

★ Same complexity as SVM in original space!